

# Exploring how LLMs could improve my work

Oleksandr Koshchii

JCNS at MLZ, Forschungszentrum Jülich GmbH, Germany

# Exploring existing models

- Lots of models:
  - Anthropic: Claude,
  - DeepSeek: V3/R1,
  - Google: Gemini,
  - Magistral: Mistral,
  - Meta: Llama,
  - Microsoft: Phi,
  - OpenAI: GPT-5,
  - xAI: Grok,
  - and many more...  
<https://huggingface.co/>
- Features:
  - Some are private (e.g. GPT-5 and Claude 4). The source code, training strategies, model weights, and even the number of parameters are not publicly available. You can't just download and run them on your own server.
  - Some are open: Llama 4, DeepSeek R1, Mistral
  - Many (hundreds) billions of parameters
  - Various improvement strategies:
    - Model scaling
    - Dedicated (targeted) training
  - Developing rapidly

## How they work

- Understanding how LLMs work is crucial.
  - Tokenization: predict likely text rather than retrieve facts (fake citations)
  - Training time cutoff limitation (tax law changed, policy has been updated)
  - Using new chat window can speed up the process and improve quality of the answers
    - Faster responses (because the model processes less text)
    - Clears context “noise” (confusion about what you currently want)
- Dedicated prompt with context is much more useful:
  - Tailored and relevant responses
  - More accurate answers
- Always assess feasibility of the responses from LLMs

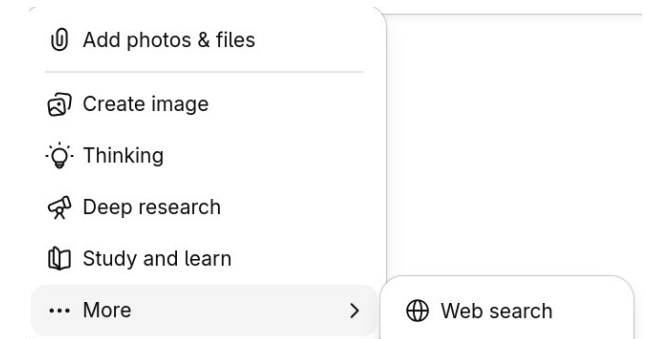
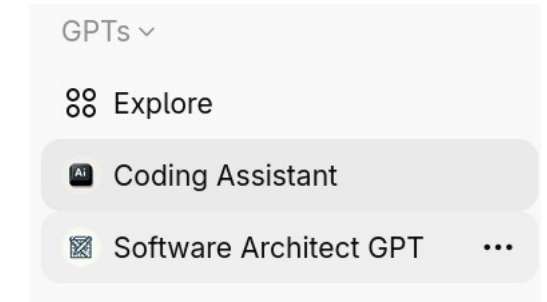


## Using coding assistants

- Github Copilot – auto-complete scaled on steroids:
  - LLM that has access to the whole repo/project.
  - Free tier: up to 2,000 code completions per month within supported IDEs (80 per working day), up to 50 premium chat requests per month, access to advanced AI models, including GPT-4o and Claude 3.5 Sonnet, for generating code and providing explanations.
  - Code completion, boilerplate generation, speeding up typical developer tasks (writing functions, classes, documentation, code refactoring) are well supported.
  - Chat+ context features: support of slash commands like `/explain`, `/doc`, `/fix`, `/tests`; inline chats with dedicated scope like `@vscode`, `@terminal`, `@workspace`, etc.
- “Heavier” solutions are also available, e. g. Claude Code.
- Certain tasks can be delegated to AI Agents. Example: Amazon’s upgrade of Java applications from v8, v11 to v17 to dedicated agent: claim to have saved 4500 developer years.

## What else

- Paid plans provide access to more features. E.g. ChatGPT+:
  - You can create a custom GPT with built-in prompts or settings
  - More advanced models and faster processing times of requests
  - Codex agent for coding
  - OpenAI browser
- Integration throughout the entire computer ecosystem is inevitable
- Play with new features to keep up with the changes



Q&A